Problem B. Even Modulo Pair

Time Limit 1000 ms
Mem Limit 262144 kB

You are given a **strictly increasing** sequence of positive integers $a_1 < a_2 < \ldots < a_n$. Find two distinct elements x and y from the sequence such that $x < y$ and $y \bmod x$ is even, or determine that no such pair exists.

$p \bmod q$ denotes the remainder from dividing p by q.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 2 \cdot 10^4$). The description of the test cases follows.

The first line of each test case contains one integer n ($2 \le n \le 10^5$) — the length of the sequence.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_1 < \ldots < a_n \le 10^9$) — the given sequence.

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Output

For each test case:

- If no such pair exists, output `-1`.
- Otherwise, output two integers x and y — the elements that satisfy the condition.

If there are multiple valid pairs, you may output any of them.

Examples

Input	Output
4 5 1 3 4 5 6 6 2 3 5 7 11 13 4 2 3 13 37 3 17 117 1117	3 5 3 11 -1 17 1117

Note

Visualizer link

In the first test case, choosing $x = 3$ and $y = 5$ yields $y \bmod x = 5 \bmod 3 = 2$, which is even.

In the third test case, it is clear that no valid pair exists.